# The 1D advection-diffusion equation

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Consider a scalar quantity $f(x,\ t)$ such as concentration of dissolved species in a liquid, or temperature of a fluid, that is governed by the one-dimensional advection-diffusion equation

$$\frac{\partial f}{\partial t} + U\frac{\partial f}{\partial x} = \alpha\frac{\partial^2 f}{\partial x^2}, \tag{1}$$

subject to the initial condition

$$f(x,\ t=0) = \mathrm{sech}^2[k(x+2)] \tag{2}$$

on the domain $x = [-10,\ 10]$ and $0 \leq t \leq 20$. Note that equations (1) and (2) above are assumed to be dimensionless, meaning that the spatial coordinate $x$, time coordinate $t$, velocity $U$ and diffusivity $\alpha$ are all dimensionless quantities.

For instance, the advection-diffusion equation implies that the concentration of species $f$ changes in time because species are transported by advection (carried by the flow velocity $U$) and by diffusion (species diffuse to smooth out any concentration gradients, at a speed determined by the diffusivity $\alpha$).

We will solve the system above imposing (i) Dirichet boundary conditions,

$$f(x=-10,\ t) = 0, \quad f(x=10,\ t) = 0, \tag{3}$$

then (ii) Neumann boundary conditions,

$$\frac{\partial f}{\partial x}(-10,\ t) = 0, \quad \frac{\partial f}{\partial x}(10,\ t) = 0, \tag{4}$$

and finally (iii) periodic boundary conditions,

$$\frac{\partial f}{\partial x}(-10,\ t) = \frac{\partial f}{\partial x}(10,\ t). \tag{5}$$

We will employ a finite-differences FTCS scheme, i.e., forward in time, centered in space. To use this scheme, the timestep $\Delta t$ must be sufficiently small so that the stability conditions

$$\frac{U\Delta t}{\Delta x} \leq \frac{1}{2}, \quad \frac{\alpha\Delta t}{\Delta x^2} \leq \frac{1}{2} \tag{6}$$

are satisfied.

Making use of the following finite difference approximations

$$\frac{\partial f}{\partial t}(x_i, t_{n+1}) = \frac{f(x_i, t_{n+1}) - f(x_i,\ t_n)}{\Delta t} + \mathscr{O}(\Delta t) \tag{7}$$

$$\frac{\partial f}{\partial x}(x_i,\ t_n) = \frac{f(x_{i+1},\ t_n) - f(x_{i-1},\ t_n)}{2\Delta x} + \mathscr{O}((\Delta x)^2) \tag{8}$$

$$\frac{\partial^2 f}{\partial x^2}(x_i,\ t_n) = \frac{f(x_{i+1},\ t_n) - 2f(x_i,\ t_n) + f(x_{i-1},\ t_n)}{(\Delta x)^2} + \mathscr{O}((\Delta x)^2), \tag{9}$$

we can discretise equation (1) as

$$f(x_i,\ t_{n+1}) = f(x_i,\ t_n) - U\Delta t\left[\frac{f(x_{i+1},\ t_n) - f(x_{i-1},\ t_n)}{2\Delta x}\right] + \alpha\Delta t\left[\frac{f(x_{i+1},\ t_n) - 2f(x_i,\ t_n) + f(x_{i-1},\ t_n)}{(\Delta x)^2}\right]. \tag{10}$$

# Assignment

1. Open and run on Matlab the script Lab1AdvDiffEq.m. The default values should be set to $k = 0.5$, $U = 1$ and $\alpha = 1$.

2. Making use of equation (10), compute `f(i, n+1)` for the inner points, i.e., the points whose index $i$ lies within 2 and $Nx - 1$. Run the simulation with Dirichlet boundary conditions.

3. Code expressions for `f(1, n+1)` and `f(Nx, n+1)` for the points at the boundary for 'Neumann' and 'Periodic' boundary conditions and run the simulation.

   To do so, one may notice from equation (10) that we need $f(x_0,\, t_n)$ to compute $f(x_1,\, t_{n+1})$ and we need $f(x_{Nx+1},\, t_n)$ to compute $f(x_{Nx},\, t_{n+1})$. However, index $i$ runs from 1 to $Nx$, so nodes 0 and $Nx+1$ do not exist (i.e., $x_0$ and $x_{Nx+1}$ do not exist). We can overcome this difficulty by noting that the Neumann boundary conditions (1st order derivative is zero) imply that

   $$f(x_0,\, t_n) = f(x_2,\, t_n), \quad f(x_{Nx+1},\, t_n) = f(x_{Nx-1},\, t_n). \tag{11}$$

   Similarly, for periodic boundary conditions, we have that

   $$f(x_0,\, t_n) = f(x_{Nx-1},\, t_n), \quad f(x_{Nx+1},\, t_n) = f(x_2,\, t_n) \tag{12}$$

4. Play with the simulation. Change the values of $\alpha$ and $U$ in the range 0.1 to 10 (or more!) What effect dominates when $U \gg \alpha$? And when $U \ll \alpha$?

5. **Report**: the report should consist only of two independent graphs of $f$ vs. $x$. The first graph must show the solution obtained with Neumann boundary conditions. The second graph must show the solution obtained when using Periodic boundary conditions. Each graph should contain 3 curves for $f(x,\, t)$ evaluated at $t = 1$, $t = 5$ and $t = 15$. You can set $\alpha$ and $U$ to different values of your choosing, and make sure to indicate on the figure captions the values that you are using.

   The report, or rather the two captioned graphs, cannot extend longer than a single A4 page overall. Make sure to properly label and caption your graphs.

   You may find it useful to know that typing `index = find(t>=5, 1)` in Matlab gives the first time index for which $t \geq 5$.